

NLP A3

How LLMs Handle Ambiguity Across Multiple Languages

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1 Motivation

Large Language Models (LLMs) have demonstrated strong performance across a wide range of natural language processing tasks. However, their behavior in the presence of ambiguity remains less understood. Natural language is inherently ambiguous, and human speakers rely on context, shared knowledge, and pragmatic reasoning to resolve such ambiguity.

In contrast, LLMs are often expected to provide precise answers even when questions lack sufficient context. This raises concerns about potential biases, overconfidence, and reliability, especially when LLMs are used in real-world applications. Recent work has also highlighted that LLMs tend to exhibit an English-centric bias, performing less reliably in other languages (Guo et al., 2025; Venkateswaran et al., 2026).

This paper investigates how several open-source LLMs handle ambiguous and underspecified questions across different prompting strategies and languages (English and Dutch). The goal is to explore whether LLMs appropriately reflect ambiguity, how often they fail, and to what extent their behavior is influenced by language and prompting style.

2 Relevance

Several core challenges in linguistics and natural language processing are addressed, particularly in relation to ambiguity, cross-linguistic variation, and LLM behavior.

2.1 Ambiguity

Ambiguity is a fundamental property of natural language and has been widely studied in semantics and pragmatics (Fortuny and Payrató, 2024). Words and phrases can have multiple meanings depending on context, requiring speakers to rely on world knowledge and inference to resolve interpretation. Questions such as "Where is Cambridge?"

or "What is Mercury?" are inherently ambiguous and have multiple valid interpretations.

Recent work has shown that LLMs often struggle with ambiguity (Keluskar et al., 2024; Zhou et al., 2023), tending to produce a single, overconfident answer rather than explicitly acknowledging that there could be multiple interpretations. This behavior reflects the limitations of LLMs and how they approach meaning without true understanding.

2.2 Linguistic Variation

LLMs are predominantly trained on English data, which can lead to performance disparities across different languages. Previous research has shown that there is a bias towards the English language in multilingual models, and performance degrades for languages that are represented less in the training data (Bender et al., 2021; Terryn and De Lhoneux, 2024). By comparing the LLM outputs for the questions in both English and Dutch, this study investigates whether such a bias is observable in ambiguity resolution, and whether models fail more frequently or behave differently when operating in a non-English context.

2.3 Semantics and Heuristics

LLMs do not explicitly model meaning in the same way as traditional semantic frameworks. Instead, they rely on statistical patterns learned from large datasets. As a result, their responses to ambiguous input may reflect heuristic strategies rather than true interpretation. (Kintsch and Mangalath, 2011)

In this study, model outputs are manually categorized into different answer types (single, multiple, generic, uncertain, failure), which provides insight into how the models handle the ambiguous input.

3 Methodology

3.1 Models

Three open-source LLMs were used: LLaMa (8B), Mistral (7B), and BLOOM (3B). These models were chosen to compare LLMs of different sizes, with different training setups, and varying multilingual capabilities. All models are queried using a temperature of 0.7 with sampling enabled, and a fixed maximum token length. This allows for some variability in responses while maintaining comparability across models.

3.2 Dataset

A dataset of 30 ambiguous questions was manually created and then translated into Dutch. Translations were kept as literal as possible to preserve ambiguity across languages. The goal was to include queries that allow for multiple valid interpretations without additional context. The questions cover:

- Locations: e.g. "Where is Cambridge?"
- Entities/objects: e.g. "What is Mercury?"
- Roles/names: e.g. "Who is Jordan?"

3.3 Prompting Strategies

Every question was tested with 3 prompt types:

1. Basic: base question without instructions
2. Instruction: "Answer the following question as precisely as possible: ..."
3. Forced: "If multiple interpretations exist, choose the most likely one. ..."

These prompting strategies were selected to test how model behavior changes when ambiguity is left open or when it is constrained. No additional fine-tuning or prompt engineering beyond the defined strategies was applied.

3.4 Evaluation Strategy

Each response was manually labeled into one answer type:

- Single: one interpretation chosen;
- Multiple: multiple interpretations mentioned;
- Generic: vague or generic explanation;
- Uncertain: model expresses uncertainty;

- Failure: irrelevant, incoherent, or broken output.

Each model generated a response for each prompt type for each of the 30 questions in both languages, resulting in 180 outputs per model and 530 outputs in total.

4 Evaluation

4.1 Model Comparison

Table 1: Answer Types Across LLMs

Model	failure	generic	multiple	single	uncertain	Grand Total
<i>bloom</i>	88	19	1	60	2	170
<i>llama</i>	2	19	16	143		180
<i>mistral</i>	4	12	8	136	20	180
Grand Total	94	50	25	339	22	530

The results show a clear difference in reliability across models. BLOOM produces a disproportionately high number of failures (88 out of 94 total failures, or 93.6%), indicating instability and poor handling of ambiguous queries. In contrast, LLaMa and Mistral both produce very few failures and a large number of single-interpretation answers, suggesting higher robustness.

4.2 Language Comparison

Table 2: Answer Types Across Languages

Language	failure	generic	multiple	single	uncertain	Grand Total
<i>en</i>	38	31	13	167	14	263
<i>nl</i>	56	19	12	172	8	267
Grand Total	94	50	25	339	22	530

Across languages, Dutch prompts result in more failures (56) than English prompts (38). This suggests a degree of language bias, where models perform less reliably in Dutch. However, this effect is primarily driven by BLOOM, while LLaMa and Mistral remain relatively stable across both languages. Additionally, more generic answers were given in English (31) than in Dutch (19), which may have contributed to the higher failure rate for the Dutch language. Some models seem to opt for a confident but wrong answer rather than a safe and generic answer.

4.3 Prompting Effects

Table 3: Answer Types Across Prompting Strategies

Prompt Type	failure	generic	multiple	single	uncertain	Grand Total
<i>basic</i>	30	12	5	122	6	175
<i>forced</i>	32	15	16	109	4	176
<i>instruction</i>	32	23	4	108	12	179
Grand Total	94	50	25	339	22	530

Prompting strategies do not significantly reduce failure rates, as all three prompt types result in a similar number of failures. However, forced prompting increases the number of multiple-interpretation responses rather than decreasing it, indicating that prompting can influence how ambiguity is expressed, even if it does not improve overall reliability.

Interestingly, some responses were not only influenced by prompt type in terms of structure, but also in terms of semantic interpretation. In several cases, different prompting strategies led the same model to select entirely different meanings for an ambiguous term.

For instance, LLaMa answered the Dutch question "Wat is Shell?" differently for each prompt type: a biological meaning (a protective outer layer), a corporate entity (the oil company), and a technical meaning (a network related term). This behavior indicates that LLMs do not maintain a stable internal preference for a single interpretation in ambiguous cases. Instead, their outputs are highly sensitive to prompt formulation, which raises concerns about consistency and reliability in real-life applications.

4.4 Ambiguity Handling

Table 4: Failures Across LLMs

Model	en	nl
<i>bloom</i>	36	52
<i>llama</i>		2
<i>mistral</i>	2	2
Grand Total	38	56

Failure analysis reveals that BLOOM is particularly sensitive to language, with substantially more failures in Dutch than in English. LLaMa and Mistral show minimal differences, suggesting that larger models are more robust across languages. The results indicate that LLMs struggle to appropriately handle ambiguity, often resolving underspecified questions by selecting single interpretations rather than acknowledging uncertainty.

5 Final Thoughts

According to the results, LLMs are highly sensitive to ambiguity, often resolving it incorrectly. There is some evidence of language bias, especially in a smaller model like BLOOM. Using different prompting strategies does change behavior, but it

does not fix reliability issues. Lastly, models tend to guess or choose one interpretation rather than admit uncertainty.

Some of the limitations of the study are: manual labeling could introduce some subjectivity; the size of the dataset is limited; and there is no fine-grained evaluation of output correctness.

In the future, this work could be expanded by using a larger dataset and by testing more languages and models to see how the results change. Additionally, analyzing the output and which interpretation is chosen could reveal further biases and insights.

References

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A Appendix

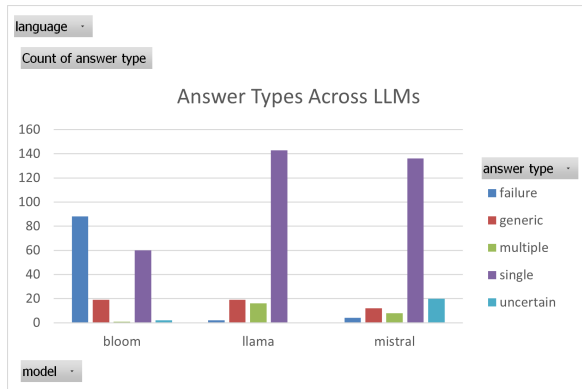


Figure 1: Answer Types Across LLMs.

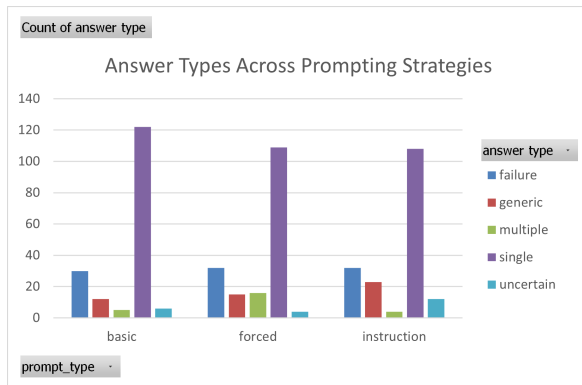


Figure 2: Answer Types Across Prompting Strategies.

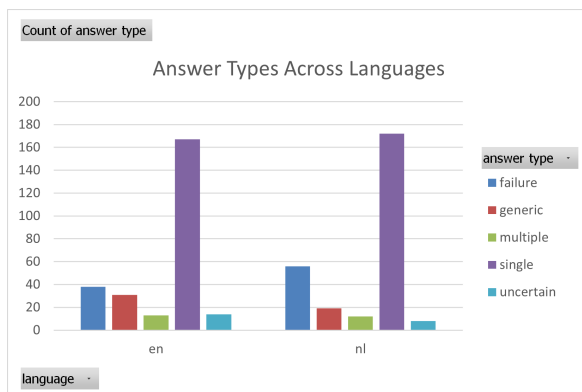


Figure 3: Answer Types Across Languages.

A.1 English Questions

1. "Where is Cambridge?",
2. "Where is Springfield?",
3. "Where is Victoria?",
4. "Where is Oxford?",

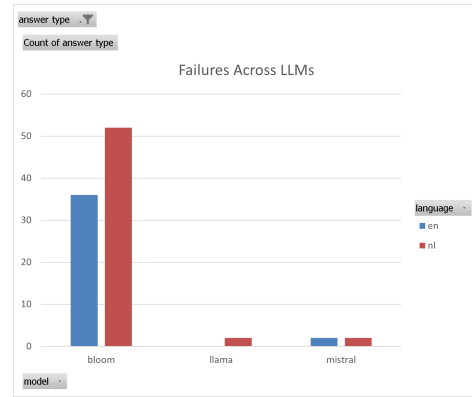


Figure 4: Failures Across LLMs.

5. "Where is San José?",
6. "Where is Quebec?",
7. "Where is Perth?",
8. "Where is Kingston?",
9. "Where is Melbourne?",
10. "Where is Newcastle?",
11. "What is Apple?",
12. "What is Amazon?",
13. "What is Java?",
14. "What is Python?",
15. "What is Tesla?",
16. "What is Mercury?",
17. "What is Windows?",
18. "What is Oracle?",
19. "What is Shell?",
20. "What is Delta?",
21. "Who is Jordan?",
22. "Who is Taylor?",
23. "Who is Morgan?",
24. "Who is Alex?",
25. "Who is Kim?",
26. "Who is the president?",
27. "Who is the prime minister?",

28. "Who is the king?",
29. "Who is the CEO?",
30. "Who is the leader of the country?"

A.2 Dutch Questions

1. "Waar ligt Cambridge?",
2. "Waar ligt Springfield?",
3. "Waar ligt Victoria?",
4. "Waar ligt Oxford?",
5. "Waar ligt San José?",
6. "Waar ligt Quebec?",
7. "Waar ligt Perth?",
8. "Waar ligt Kingston?",
9. "Waar ligt Melbourne?",
10. "Waar ligt Newcastle?",
11. "Wat is Apple?",
12. "Wat is Amazon?",
13. "Wat is Java?",
14. "Wat is Python?",
15. "Wat is Tesla?",
16. "Wat is Mercury?",
17. "Wat is Windows?",
18. "Wat is Oracle?",
19. "Wat is Shell?",
20. "Wat is Delta?",
21. "Wie is Jordan?",
22. "Wie is Taylor?",
23. "Wie is Morgan?",
24. "Wie is Alex?",
25. "Wie is Kim?",
26. "Wie is de president?",
27. "Wie is de premier?",
28. "Wie is de koning?",
29. "Wie is de CEO?",
30. "Wie is de leider van het land?"